

Fig. 4: Actual-size PCB layout for the power supply

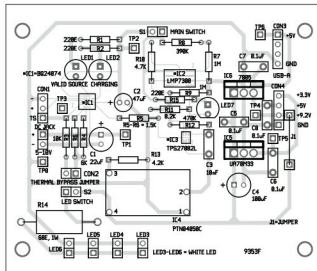


Fig. 5: Component layout for the PCB

circuit is shown Fig. 4 and its component layout in Fig. 5. IC1, IC2 and IC3 are SMD components which are to be mounted on the solder side of the PCB. Suitable heat sinks should be used for IC5 and IC6. Connect the Li-ion battery at CON1 and set the thermal bypass jumper at CON2 in the circuit, as explained under 'battery management' section.

Check that the voltages at various points in the circuit are as per test point table before powering it for the first time. Enclose the power supply circuit in a suitable box. Fix DC jack, CON1 through CON4, switches S1 and S2 appropriately on the box. Fix LED3 through LED6 such that you can use

them conveniently for torchlight.

On switching S1 on the power supply should provide 9.2V, 5V and 3.3V at the output terminals (CON4). It is safe to draw about 800mA from the 9.2V and 5V terminals while 3.3V terminal is able to provide 500mA of current only. Switch S2 turns on the torchlight (provided switch S1 is on).

You can connect Li-ion battery up to 3000mAh capacity to the circuit but you have to recalculate the current-setting resistors as described in the previous section. When the battery voltage reaches below 2.8V, LED7 glows to indicate the 'end of capacity' of the battery and the loads are automatically turned off.

To recharge the battery, you can connect the circuit to an adaptor producing anywhere from 5V to 10V DC. You can also charge the battery by connecting it to a solar panel providing anywhere from 5V to 10V. If the charge input source is a valid source, green LED1 glows. LED2 glows to indicate that the battery is being charged and goes off after the battery is fully charged. LED2 also flashes at the rate of 2Hz if the IC's internal safety timers have expired, an error state which indicates that your battery is not in good condition. ●

The author is a B.Tech in electronics and instrumentation. His interests include embedded systems, software programming and instrumentation



Spot an Error

And WIN Award!

You can win one thousand rupees plus a certificate from the editor of Electronics For You magazine if you spot and report any factual error in the latest issue of the magazine. Your report will be published in a coming issue of the magazine, giving credit to you.

Though every care is taken in preparation of the content for the magazine, an error can slip in at times. So we invite our learned readers to point out any error that they spot, for the benefit of the other readers.

Rules:

- The error has to be reported before end of the month that is mentioned on cover of the magazine
- The report has to be sent in writing or by email addressed to the Editor, EFY Magazine (editsec@efyindia.com)
- The error should be of factual nature, in the information published in the magazine, and not a grammatical or syntax error.
- The correct information should also be provided along with the error spotted by you
- In case more than one person reports the same error, the one whose report reaches us the earliest will get the award money
- In case two or more persons jointly report an error, the award money will be divided amongst them
- EFY employees, their family members and friends, and EFY associates and their family members and friends are not eligible
- Decision of the editor will be final in this matter

electronics
FOR YOU